8.1 Activities: fractions, decimals and percentages

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Each activity takes 10–20 minutes. Do them at the kitchen table, out loud, with no writing pressure. If an activity goes well, stop while it is still fun and repeat it another day.

1. The euro grid

You need: one €1 coin (or a drawn circle labelled €1), a handful of 1-cent, 5-cent, 10-cent, 20-cent and 50-cent coins (real or paper).

One euro is 100 cents, which makes it a perfect "100%". Lay out the €1. Ask him to show half a euro in coins: 50 cents. Write it three ways together:

Fraction	Decimal	Percentage	Coins
½	€0,50	50%	50c
¼	€0,25	25%	20c + 5c
¾	€0,75	75%	50c + 20c + 5c
1/10	€0,10	10%	10c
1/5	€0,20	20%	20c
1/100	€0,01	1%	1c

Let him make each one with coins before writing it. Then reverse: you lay down coins (say 60 cents) and he names all three outfits (3/5, 0,6, 60%).

Talk about it: "Why do you think 'per cent' means 'out of a hundred'?" (*Cent* comes from the Latin for hundred, just like a euro has 100 cents.)

2. Three-outfits matching game

You need: 18 small cards (cut-up index cards).

Write six numbers three ways each: ½ / 0,5 / 50%, ¼ / 0,25 / 25%, ¾ / 0,75 / 75%, 1/10 / 0,1 / 10%, 1/5 / 0,2 / 20%, 2/5 / 0,4 / 40%. Shuffle and lay face down. Play memory, but a match is any **three** cards that are the same number (turn over three per turn). Collect sets.

Make it easier by playing face up and just sorting into families. Make it harder later by adding 1/8 / 0,125 / 12,5% and 3/10 / 0,3 / 30%.

Talk about it: "Which outfit is easiest for you to picture? Which is the trickiest?"

3. The washing line

You need: a piece of string tied between two chairs, pegs or paperclips, cards.

Write 0 at one end and 1 (= 100%) at the other. Hand him cards one at a time: 0,5, 25%, ¾, 0,1, 90%, 1/3, 0,35, 2/5. He pegs each on the line where it belongs. Ask him to explain one placement each round. Move cards that are wrong together ("Hmm, is 0,35 nearer to ¼ or to ½?").

Talk about it: "Which two were hardest to put in order? How did you decide?"

4. Supermarket folder hunt

You need: the weekly supermarket leaflet (*folder*) from the letterbox or an app, a pencil.

Hunt for every %, “1 + 1 gratis”, “2e halve prijs” and “van … voor …” deal. For each one, ask: “How much do you actually save? What do you pay?” Start with the 50% offers (halve the price), then 25%, then the trickier ones.

A “van €4,00 voor €3,00” deal is a nice reverse puzzle: €1 off out of €4 is ¼, so 25% korting.

Talk about it: “Is ‘1 + 1 gratis’ the same as 50% korting? When is it a good deal and when isn’t it?” (It’s the same as 50% off each item if you’d use both. If you only need one, you save nothing.)

5. Shop with korting and btw

You need: 6–8 household items, sticky notes, play money or paper.

He runs a shop. Write a price on each item (whole euros, such as €20, €40, €60, €80). Then you announce a sale: “Today everything is 25% off!” He makes new price labels. Swap roles. Later add a “shopkeeper’s pricing” round: the price on the label is “excl. btw” and he has to add 21% for the customer (10% + 10% + 1%).

Talk about it: “Who pays the btw: the shop or you? Where does the money go?” (The customer pays it as part of the price; the shop passes it on to the government.)

6. Percentage of the day

You need: a sheet of paper and coloured pencils.

Draw a bar for 24 hours. Together estimate how many hours he sleeps, is at school, plays, eats. Shade them. Then turn them into rough percentages: 8 hours of sleep is 1/3 of the day, about 33%. 6 hours of school is ¼ of the day, 25%.

Talk about it: “Does the bar add up to 100%? Why must it?”

Conversation prompts for any day

- “The sign says 30% korting. Is that more or less than a third off?”
- “My phone battery is at 40%. About what fraction is that?”
- “If you got 18 out of 20 on a test, what percentage would that be? Is that good?”
- “Would you rather have ¾ of a pizza or 70% of the same pizza?”
- “What’s something that’s 100% true about you?”